

Experiment 2: Missing Value Analysis and Outlier Detection

Aim

To analyze missing values in a dataset, perform appropriate imputation techniques, and detect outliers using statistical methods such as IQR, Z-score, and percentile approach.

Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import zscore

df = pd.read_csv("insurance.csv")

np.random.seed(42)

for col in ['age', 'bmi', 'charges']:
    df.loc[df.sample(frac=0.1).index, col] = np.nan

df.loc[df.sample(frac=0.1).index, 'sex'] = np.nan

print("Missing Values:\n", df.isnull().sum())
print("\nStatistical Summary:\n", df.describe())

sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title('Missing Value Heatmap')
plt.show()

df_impute = df.copy()

df_impute['age'].fillna(df_impute['age'].mean(), inplace=True)
df_impute['bmi'].fillna(df_impute['bmi'].median(), inplace=True)
df_impute['charges'].fillna(df_impute['charges'].mean(), inplace=True)
df_impute['sex'].fillna(df_impute['sex'].mode()[0], inplace=True)

sns.boxplot(x=df_impute['charges'])
plt.title('Charges Boxplot')
plt.show()

Q1 = df_impute['charges'].quantile(0.25)
Q3 = df_impute['charges'].quantile(0.75)
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
```

```

iqr_outliers = df_impute[
    (df_impute['charges'] < lower) | (df_impute['charges'] > upper)
]

z_outliers = df_impute[
    np.abs(zscore(df_impute['charges'])) > 3
]

lower_p = df_impute['charges'].quantile(0.01)
upper_p = df_impute['charges'].quantile(0.99)

percentile_outliers = df_impute[
    (df_impute['charges'] < lower_p) | (df_impute['charges'] > upper_p)
]

print("IQR Outliers:", len(iqr_outliers))
print("Z-score Outliers:", len(z_outliers))
print("Percentile Outliers:", len(percentile_outliers))

```

Output

Missing Values:

```

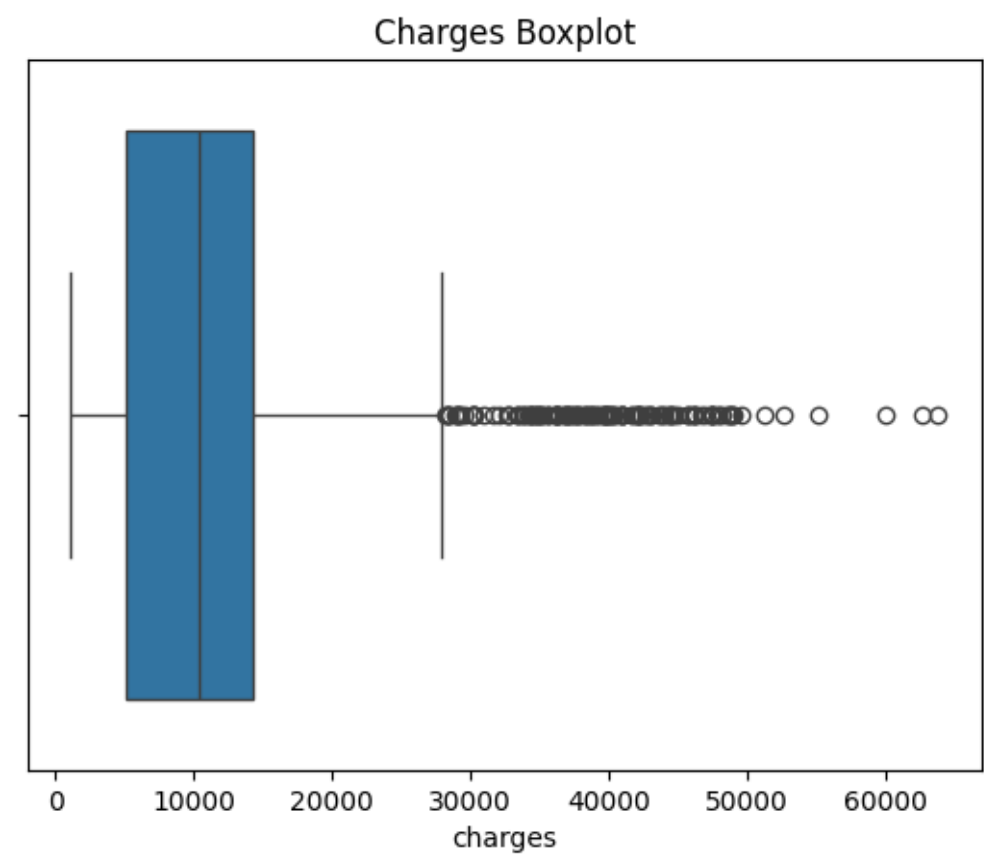
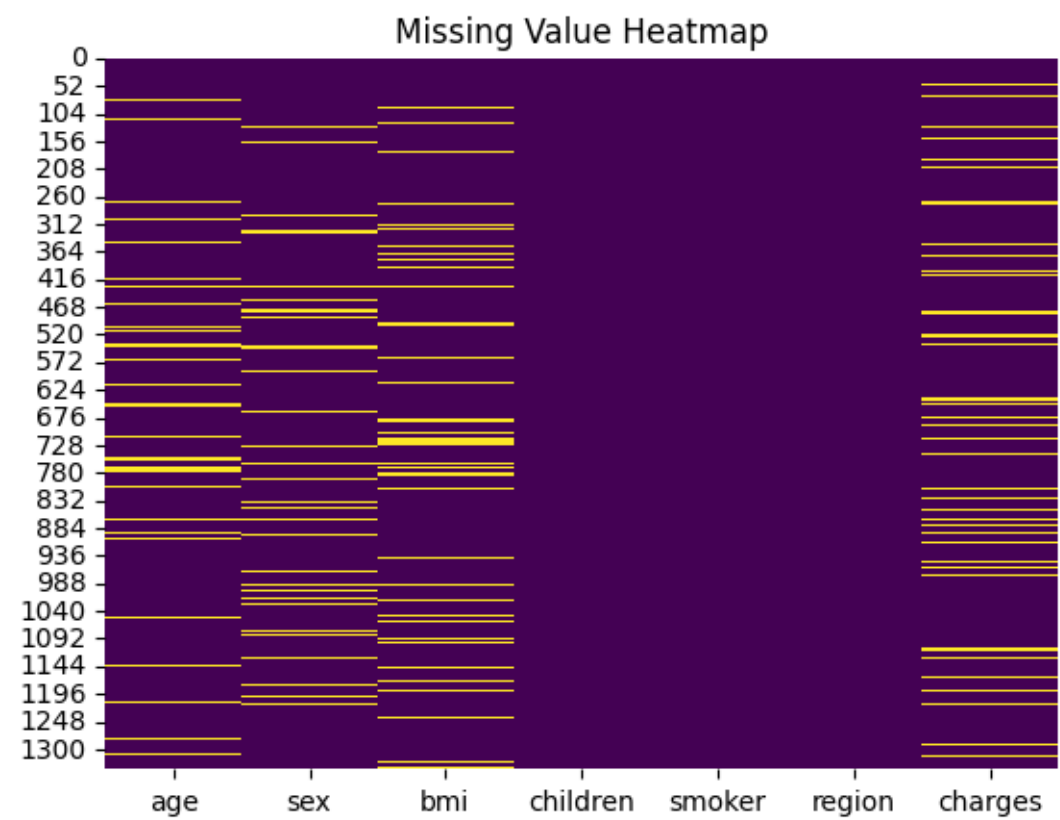
age      134
sex      134
bmi      134
children  0
smoker    0
region    0
charges  134
dtype: int64

```

Statistical Summary:

	age	bmi	children	charges
count	1204.000000	1204.000000	1338.000000	1204.000000
mean	39.206811	30.632566	1.094918	13056.041441
std	14.080740	6.159517	1.205493	11831.721806
min	18.000000	15.960000	0.000000	1121.873900
25%	26.000000	26.195000	0.000000	4687.444925
50%	39.000000	30.250000	1.000000	9285.794350

75% 51.000000 34.777500 2.000000 16579.959053
max 64.000000 53.130000 5.000000 63770.428010



IQR Outliers: 149

Z-score Outliers: 22

Percentile Outliers: 28

Result

Missing values in the dataset were successfully identified and handled using mean, median, and mode imputation techniques. Visualization using a heatmap clearly showed the distribution of missing data. Outliers in the 'charges' column were detected using three methods: IQR, Z-score, and percentile approach. The number of outliers identified varied across methods, demonstrating the effectiveness of multiple statistical techniques for robust outlier detection.